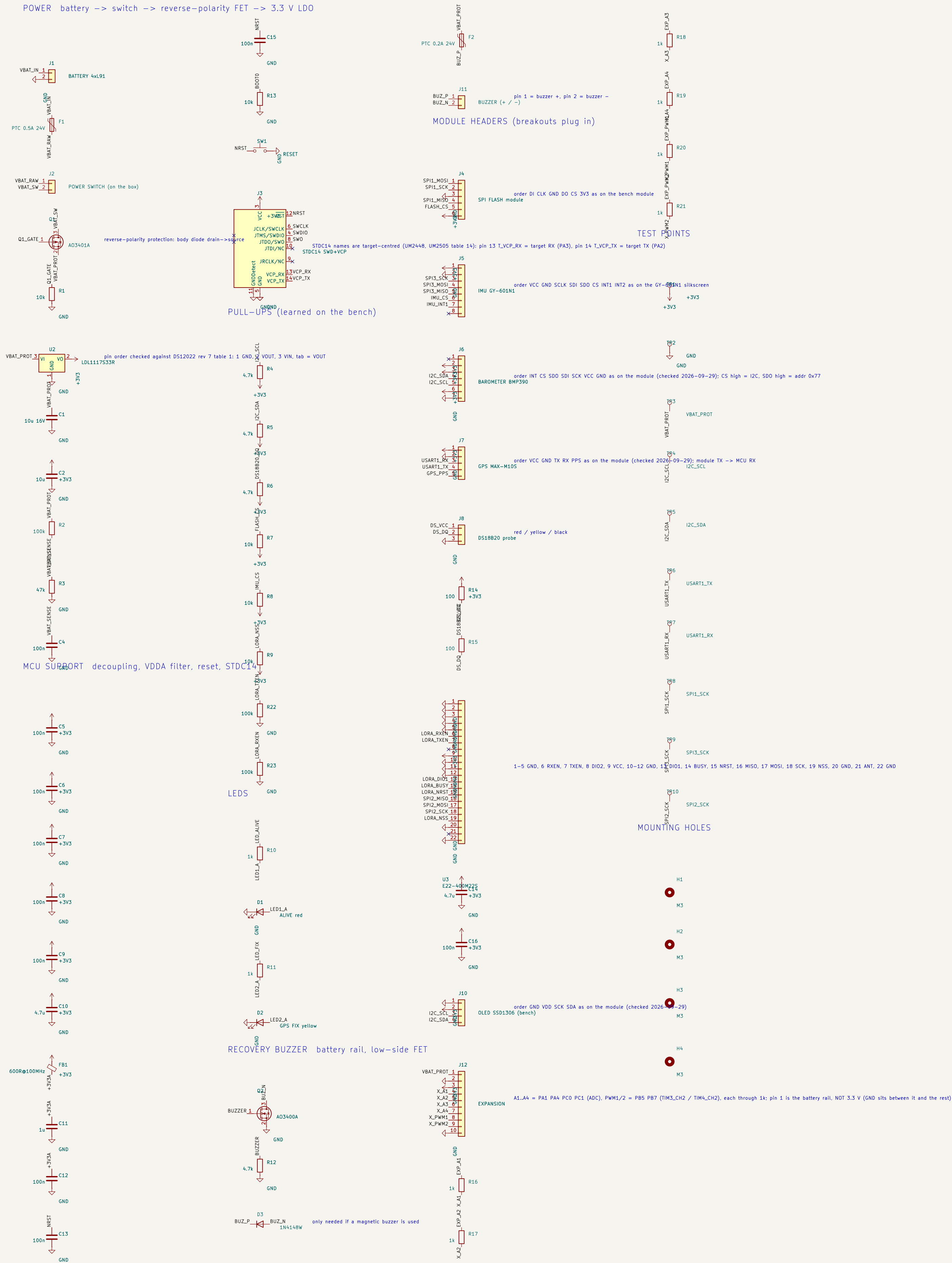
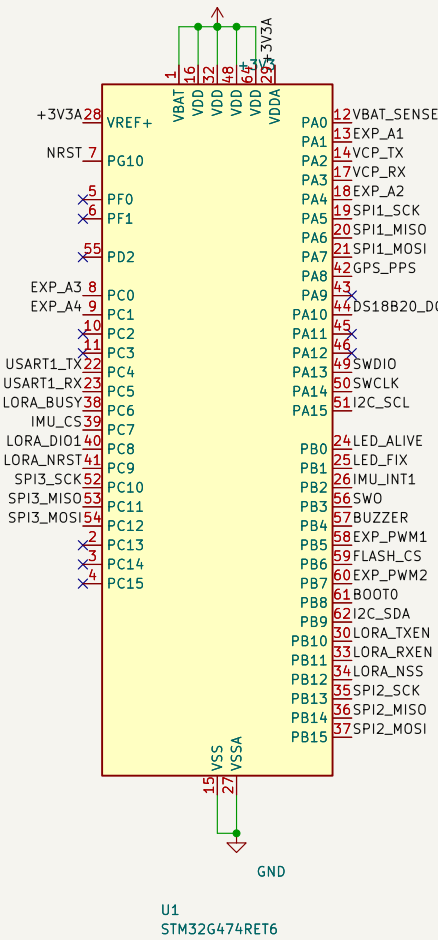
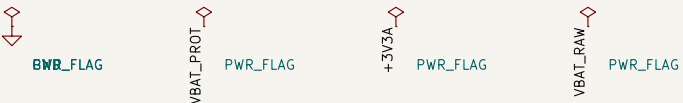


MCU STM32G474RET6 (same chip and pins as the NUCLEO-G474RE bench)

POWER battery -> switch -> reverse-polarity FET -> 3.3 V LDO



POWER FLAGS (for ERC)



RECOVERY BUZZER battery rail, low-side FET

EXPANSION A1..A4 = PA1 PAA PC0 PC1 (ADC), PAW1/2 = PB5 PB7 (TIM3_CH2 / TIM4_CH2), each through 1k; pin 1 is the battery rail, NOT 3.3 V (GND sits between it and the rest)